TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Moisture Regain and Moisture Content of Cotton, Fibre Length, Fineness, Maturity Ratio and Trash Content

Five determinations that describe a cotton lot before it is spun: how much water it holds, how long and how fine its fibres are, how mature they are, and how much of the bale is not fibre at all.

TE-3104 · QUALITY EVALUATION OF FIBRE AND YARN SESSIONAL · EXPERIMENT 1 · LEVEL 3 TERM 1

1. Aim and theory

Aim: to determine, on a representative sample of raw cotton, (a) moisture content and moisture regain by the oven-dry method, (b) fibre length and its distribution, (c) fibre fineness by the micronaire (airflow) method, (d) maturity ratio by caustic-soda swelling, and (e) trash content by the Shirley analyser. Together these five results decide the commercial condition of the lot, the spinning limit it can support, and the waste that must be removed before carding.

1.1 Moisture content and moisture regain

Cotton is hygroscopic: its cellulose hydroxyl groups bind water from the atmosphere, so every cotton mass traded or tested contains water whose amount depends on the relative humidity and temperature around it. Two different ratios are used, and confusing them is the classic laboratory error. **Moisture content** expresses the water as a percentage of the total (wet) mass as received. **Moisture regain** expresses the same water as a percentage of the oven-dry mass. Regain is the figure used in textile commerce because the dry fibre is the stable reference: the standard (commercial) regain of cotton is 8.5%, and invoices are corrected to that regain so that neither buyer nor seller pays for excess water.

$$\text{Moisture content (\%)} = (W - D) / W \times 100$$

$$\text{Moisture regain (\%)} = (W - D) / D \times 100$$

where W is the mass of the sample as received and D is its oven-dry mass. The two are related by regain = content / (100 − content) × 100. At the standard regain of 8.5% the corresponding content is about 7.83%. Because strength, mass, length measurement and electrical properties all shift with

moisture, every other test in this sheet — and in the whole course — is performed after conditioning in the standard testing atmosphere of 20 ± 2 °C and 65 ± 2% relative humidity.

1.2 Fibre length

Length is the single most valuable cotton property: longer fibres give stronger, finer and more even yarns because more fibre-to-fibre contact points resist drafting forces. A cotton sample never has one length; it has a distribution, described by the mean length, the upper-half mean length (the mean of the longest half by weight, close to the commercial staple length), the 2.5% span length, and the uniformity index (mean length ÷ upper-half mean length × 100). Short fibres — typically those below 12.7 mm (0.5 in) — contribute almost nothing to strength and become fly and waste; their proportion is the short fibre content. In the teaching laboratory the distribution is obtained with a comb sorter (Baer sorter) or a fibrograph; the principle is the same: fibres are aligned, clamped, and withdrawn in length groups that are weighed to build the length histogram.

1.3 Fineness and the micronaire value

Fibre fineness is properly a linear density (mass per unit length, in mtex). For cotton it is measured indirectly by the **micronaire** method: a fixed mass of cotton (for example 3.24 g in the classic instrument) is compressed to a fixed volume and air is passed through the plug at a fixed pressure. Fine fibres present a larger total surface and greater resistance, so the airflow rate falls; coarse fibres let more air through. The scale reading — the micronaire value — is therefore an airflow index, not a direct mass measurement. As a useful approximation, for a mature cotton the linear density in mtex is about micronaire × 39.37. The caution matters: an immature cotton also reads low on micronaire because its thin-walled fibres collapse and resist airflow, so a low micronaire can mean either genuinely fine fibres (desirable) or immature fibres (a dyeing and neps hazard). Fineness must always be read together with maturity.

1.4 Maturity ratio

A cotton fibre matures by depositing secondary-wall cellulose inside the primary wall. A fully mature fibre has a thick wall and a small lumen; an immature fibre is a flat, thin-walled ribbon that tangles into neps, breaks in processing and dyes pale, producing white specks in dyed fabric. In the **caustic-soda swelling method** (18% NaOH), mature fibres swell into nearly cylindrical rods while immature fibres remain flat, twisted ribbons and dead fibres stay thin and transparent. Counting swollen (N), thin-walled (T) and dead (D) fibres under the microscope gives the maturity ratio:

$$\text{Maturity ratio } M = (N - D) / 200 + 0.70$$

where N and D are the percentages of normal (fully swollen) and dead fibres in a count of at least 400 fibres. A ratio of 1.0 corresponds to an average, fully standard maturity; cotton below about 0.70 is

seriously immature, and 0.80–0.95 is the normal commercial range.

1.5 Trash content

Trash means everything in ginned cotton that is not spinnable fibre: leaf, bract, stem, seed-coat fragments, sand and dust. Trash dulls needles and wires, raises card waste, weakens yarn and appears as dark specks in fabric. The **Shirley analyser** separates trash mechanically: the sample is fed between a feed roller and plate to a rapidly rotating licker-in covered with saw-tooth wire, whose pins tease the tufts open while a controlled air current carries the light lint forward and lets the heavy trash fall into a collection tray. Two passes are normally made and the recovered trash is weighed.

$$\text{Trash content (\%)} = \text{mass of trash} / \text{mass of sample fed} \times 100$$

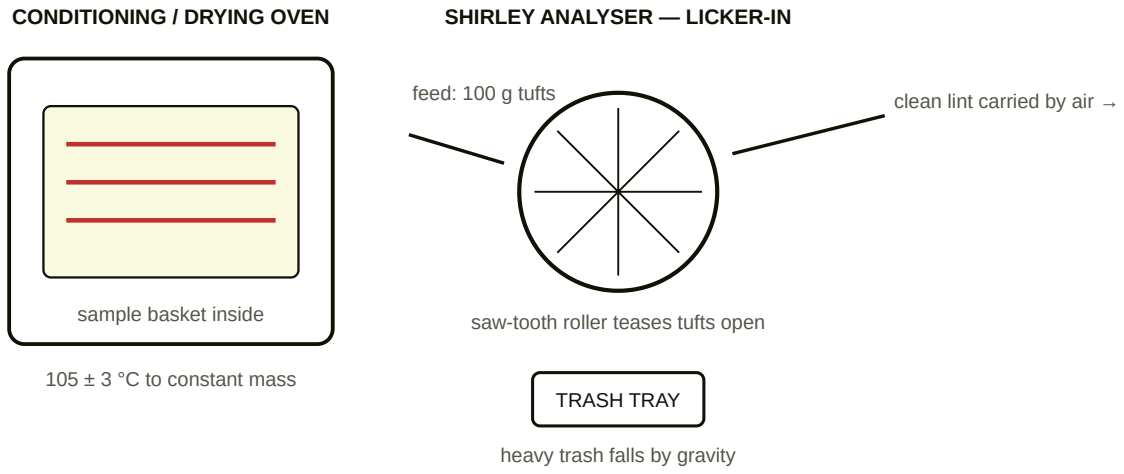


Figure 1 — Left: the ventilated drying oven used for the moisture determinations. Right: working principle of the Shirley analyser — the licker-in opens the tufts, air carries the lint onward and gravity drops the trash into the tray.

2. Apparatus and procedure

- **Apparatus:** ventilated oven (105 ± 3 °C) with weighing bottles or baskets, analytical balance (0.001 g) and laboratory balance, desiccator, conditioning room or cabinet (65 ± 2% RH, 20 ± 2 °C), comb sorter or fibrograph, micronaire instrument with compressor and calibration cottons, microscope with slides, 18% caustic soda solution, Shirley analyser, forceps and brushes.
- **Sampling and conditioning:** draw tufts from at least ten places in the lot, mix thoroughly, and condition the whole laboratory sample for at least 24 h (or 4 h with forced air movement) before any test except the as-received moisture sample, which is weighed immediately and sealed.
- **Moisture:** weigh about 50 g as received (W). Dry at 105 ± 3 °C, cooling in a desiccator before each weighing, until two successive weighings 30 min apart differ by less than 0.05% — the constant dry

mass D. Compute content and regain. Never leave the dried sample standing in room air: it begins to regain moisture within minutes.

- **Length:** prepare a representative tuft, align it in the comb sorter, withdraw successive length groups with forceps, weigh each group, and construct the length distribution; read mean length and staple (upper-half mean) length. Alternatively scan the prepared beard on the fibrograph and read the 2.5% and 50% span lengths.
- **Fineness:** weigh the exact plug mass specified for the instrument, compress it into the chamber, pass air at the set pressure and read the micronaire value. Check the instrument with two calibration cottons before and after the sample; test at least two plugs and average.
- **Maturity:** tease a small tuft into parallel fibres on a slide, flood with 18% NaOH, cover, and count at least 400 fibres under the microscope as normal (swollen, rod-like), thin-walled (flat, partly swollen) or dead (thin, unswollen). Calculate the maturity ratio from the formula above.
- **Trash:** weigh 100.0 g, feed it slowly and evenly through the Shirley analyser, collect the clean lint and the trash separately, pass the lint once more, combine the trash fractions and weigh. Account for the full mass balance — lint + trash + loss must equal the feed within experimental error.

3. Worked calculation

Moisture. Sample as received W = 50.00 g; oven-dry mass D = 46.30 g; water = 50.00 − 46.30 = 3.70 g.

$$\text{Moisture content} = 3.70 / 50.00 \times 100 = 7.40\%$$

$$\text{Moisture regain} = 3.70 / 46.30 \times 100 = 7.99\% \approx 8.0\%$$

The lot is slightly drier than the 8.5% standard regain, so its commercial (invoice) mass would be adjusted upward: correct mass = D × 1.085 = 46.30 × 1.085 = 50.23 g for every 50.00 g as received.

Fineness. Micronaire reading 4.2. Approximate linear density:

$$\text{Fineness} \approx 4.2 \times 39.37 \approx 165 \text{ mtex}$$

A value of 4.2 sits in the desirable medium-fine range (roughly 3.7–4.8 for upland cotton intended for medium counts) — provided maturity is adequate, which the next result confirms.

Maturity. Caustic-soda swelling count of 500 fibres: normal N = 68%, dead D = 8% (the remainder thin-walled).

$$M = (68 - 8) / 200 + 0.70 = 0.85$$

A maturity ratio of 0.85 is a sound, average-mature cotton: it should dye evenly and form few immature-fibre neps.

Trash. Shirley analyser, feed 100.0 g → clean lint 95.4 g, trash collected 3.9 g, unaccounted loss 0.7 g (fly and dust carried to the filter).

$$\text{Trash content} = 3.9 / 100.0 \times 100 = 3.9\%$$

A trash content of 3.9% is moderately high — typical of machine-picked cotton — and tells the blowroom to expect roughly this order of droppings before carding even begins. The mass balance closes (95.4 + 3.9 + 0.7 = 100.0 g), which validates the run.

> **READING THE FIVE RESULTS TOGETHER**
>
> Regain 8.0% (correct testing condition), micronaire 4.2 ≈ 165 mtex (medium-fine), maturity ratio 0.85 (sound), trash 3.9% (needs thorough blowroom cleaning). Fineness and maturity agree — the low-ish airflow resistance is genuine fineness, not immaturity — so this cotton can be planned for medium-count yarns with confidence.

4. Precautions

- Weigh the moisture sample immediately on receipt and keep it sealed; a few minutes of exposure in a dry laboratory changes the very quantity being measured. Dry to constant mass, cool in a desiccator, and weigh quickly.
- Condition before length, fineness and maturity testing: dry, brittle fibres break in the sorter and plug differently in the micronaire chamber.
- Draw the sample from many points of the lot. Cotton is one of the most variable natural materials; a single handful can misstate every one of the five results.
- In the micronaire test use the exact specified plug mass and verify with calibration cottons; a wrong mass shifts the airflow exactly as a wrong fineness would.
- Count at least 400 fibres for maturity, moving the slide systematically so the same fibre is never counted twice. Handle 18% NaOH as a corrosive: gloves, eye protection, no mouth pipetting.
- Feed the Shirley analyser slowly and evenly, and brush out the trash tray and roller completely between passes; trash left in the machine is trash stolen from the result.

Practice (viva preparation)

1. Distinguish moisture content from moisture regain. Which one is used commercially, and why?
2. A sample weighs 40.00 g as received and 37.40 g dry. Find its content and regain.
3. Why can a low micronaire value be either good news or bad news? What second test resolves the ambiguity?

4. What does a maturity ratio of 0.85 mean, and what happens in dyeing if the ratio falls to 0.65?
5. In the Shirley test the lint + trash weigh less than the feed. Where has the missing mass gone, and does it invalidate the trash percentage?
6. Why is the standard testing atmosphere fixed at 65% RH and 20 °C?

Answer key

1. Content = water ÷ total wet mass × 100; regain = water ÷ oven-dry mass × 100. Commerce uses regain because the dry fibre mass is invariant — a bale's fibre does not change when the weather does, only its water does, so correcting to the 8.5% standard regain prices fibre, not weather.
2. Water = 2.60 g. Content = 2.60/40.00 × 100 = 6.50%. Regain = 2.60/37.40 × 100 = 6.95%.
3. The micronaire airflow falls both for genuinely fine fibres (large surface per unit mass — desirable, more fibres in the yarn cross-section) and for immature fibres (thin walls collapse into ribbons that pack and resist air — undesirable). The caustic-soda maturity test resolves it: fine + mature is spinnable quality; fine-reading + immature predicts neps and dyeing defects.
4. It means the lot's wall development is 85% of the way described by the standard scale on which 1.0 is fully standard maturity — a sound commercial cotton. At 0.65 the fibres are seriously immature: they form neps, break in carding, and take up less dye, appearing as pale or white specks in the dyed fabric.
5. The loss is fly and fine dust drawn through the machine to the filter, plus a few fibres retained on the wire. It does not invalidate the trash result, which is defined as trash recovered ÷ feed mass; but the loss must be reported and the balance checked — a large loss (above about 1%) suggests too violent feeding and a repeat test.
6. Because cotton's mass, strength and dimensions all vary with the moisture it holds, results from different laboratories are comparable only if every sample is brought to the same equilibrium condition. 65% RH at 20 °C brings cotton close to its 8.5% standard regain, so tests are made near commercial reality.
